11·11·16 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov – Dec 2016

Max. Marks:100 | **Duration: 03 Hrs**
Class: SYBTech | Semester: III
Name of the Course: Theory of Computers | Branch: IT
Course Code: UITC306

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Differentiate deterministic and nondeterministic finite automata's. Design a DFA over $\sum = \{a, b, c\}$ that accepts the language L, such that L= {w | w is a string consisting of substrings "abc" AND "bca"} **OR** Explain subset construction method. Design a NFA over $\sum = \{0, 1\}$ that accepts the language consisting of strings starting and ending with same letter and of length >4. Convert into equivalent DFA.	12
Q 1 (b)	Minimize the given DFA by direct method (table formation) <table><tr><th></th><th>0</th><th>1</th></tr><tr><td>→A</td><td>B</td><td>E</td></tr><tr><td>B</td><td>C</td><td>F</td></tr><tr><td>* C</td><td>D</td><td>H</td></tr><tr><td>D</td><td>E</td><td>H</td></tr><tr><td>E</td><td>F</td><td>I</td></tr><tr><td>* F</td><td>G</td><td>B</td></tr><tr><td>G</td><td>H</td><td>B</td></tr><tr><td>H</td><td>I</td><td>C</td></tr><tr><td>* I</td><td>A</td><td>E</td></tr></table>	08
Q2 (a)	Construct an ε-NFA that accepts the words "man", "main" and "mains" over the alphabet of 5 letters. Convert the same into equivalent DFA. Write tuples for both the automata's.	08
Q2 (b)	Differentiate different types of FSMs with output. Construct a Mealy machine for a MOD 6 counter and convert the same into equivalent Moore machine.	12
Q3 (a)	Explain Arden's lemma. Derive regular expression for the minimized DFA of question 1(b) using Arden's lemma.	07
Q3 (b)	Give Regular expressions for the following languages consisting of the strings meeting the specified criteria and draw equivalent NFA i) Language over $\sum = \{0, 1\}$ with zero or more "01" ii) Language over $\sum = \{0, 1\}$ with no two consecutive 0's iii) Language over $\sum = \{0, 1\}$ with alternating 0's and 1's iv) Language over $\sum = \{a, b\}$ with exactly two or exactly three b's	07

Q3 (c)	Show that the language of binary strings whose length is a perfect square is not regular.	06
Q4 (a)	Convert the given grammar in GNF $G = (\{A, B, C, S\}, \{a, b, c\}, P, S)$ P: (1) $S \rightarrow ABC$ (2) $A \rightarrow aA \mid \varepsilon$ (3) $B \rightarrow bB \mid \varepsilon$ (4) $C \rightarrow cC \mid \varepsilon$ **OR** Consider (1) $S \rightarrow A1B$ (2) $A \rightarrow 0A \mid \varepsilon$ (3) $B \rightarrow 0B \mid 1B \mid \varepsilon$ i) Give leftmost and rightmost derivations for 00101 & 00011 ii) Comment on ambiguity of the grammar, justify. iii) Simplify the grammar	10
Q4 (b)	Consider the language $L = \{a^n b^m a^n \mid m, n \geq 1\}$. Construct PDA M that accepts the given language. Also construct equivalent CFG for L such that $L(G) = L(M)$ **OR** Construct a non-deterministic PDA that accepts the language of even length palindrome. Convert the same into equivalent i) Deterministic PDA ii) Context free grammar	10
Q5 (a)	Write short note on i) Church Turing Thesis ii) Variations of Turing machines	10
Q5 (b)	Define Turing Machine. Design a tur ing machine which checks for the language $L=\{a^n b^n\}$ **OR** Explain Halting Problem. Design a Turing Machine to recognize palindrome string over $\{a, b\}$	10